

Country Profile **Malawi**

Investing in Land Degradation Neutrality: Making the Case

An Overview of Indicators
and Assessments



**THE GLOBAL
MECHANISM**

United Nations Convention
to Combat Desertification

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1. Quick Facts

In Malawi, 1.3 million people were living on degrading agricultural land in 2010 – an increase of 53% in a decade, bringing the share of rural residents who inhabit degraded agricultural land up to 11% of the total rural population. Land degradation can severely influence populations' livelihood by restricting people from vital ecosystem services (including food and water), increasing the risk of poverty.

During the same time period (2000-2010), the amount of people residing in remote degrading agricultural areas with limited market access increased by 51%, reaching 207 thousand people. Populations in remote areas have restricted options for managing land and accessing other benefits of economic development.

The annual cost of land degradation in Malawi is estimated at 320 million United States dollars (USD). This is equal to 7% of the country's Gross Domestic Product. Land degradation leads to reduction in the provision of ecosystem services that takes different forms – deterioration in food availability, soil fertility, carbon sequestration capacity, wood production, groundwater recharge, etc. – with significant social and economic costs to the country.

The returns on taking action against land degradation are estimated at 3 USD for every dollar invested in restoring degraded land in Malawi. Assessments of the costs of action against land degradation through restoration and sustainable land management practices versus the cost of inaction highlight the strong economic incentive for bold actions against land degradation.

In Malawi, the Agriculture, Forestry and Other Land Use (AFOLU) sector is responsible for 78% of the total greenhouse gas emissions of the country. The removals of carbon emissions by forests are estimated at 16% of the total emissions of the country. Due to the role of terrestrial ecosystems as a source and sink of emissions land is positioned as a key point of intervention for climate change mitigation as also reflected in Malawi's Nationally Determined Contributions (NDC).

Land-based mitigation options rank among the most cost-effective opportunities to sequester carbon emissions. Economic evaluations of various climate change mitigation alternatives show that capturing carbon through restoring degraded lands (including degraded-forest reforestation) is a cost-effective option that offers multiple co-benefits.

Sustainable Development Goal 15, 'Life on Land', and its target 15.3 on Land Degradation Neutrality (LDN) is a unique opportunity for countries to curb the growing threats of land degradation and to reap multiple socioeconomic benefits of LDN. Malawi has already set national voluntary LDN targets, established the LDN baseline, and formulated associated measures to achieve LDN.

2. Population on Degrading Land

2.1 National Overview

Land is a source of well-being for present and future generations – it provides a wide range of ecosystem services that sustain human needs. Land degradation can severely influence livelihoods by limiting the availability of vital ecosystem services (including food and water), increasing the risk of poverty⁽¹⁾ and ultimately forcing people to migrate.⁽²⁾

A recent study⁽³⁾ shows that the state of the land, whether it is improving or degrading, can to a large extent influence the impact of the country's economic growth on the alleviation of poverty, making land an accelerator (or decelerator) of poverty eradication.

Poverty in Malawi is estimated to affect 57% of the rural population.⁽⁴⁾ In 2010, 11% of the rural population of the country was living on degrading agricultural land, which amounts to approximately 1.3 million people.ⁱ Moreover, between the years 2000 and 2010, the number of people living on degrading agricultural land grew by 460 thousand, representing an increase of 53% over the decade (see table 1 for further details).

By 2010, 207 thousand people or 2% of Malawi's rural population resided in remoteⁱⁱ degrading agricultural areas without market access. This number increased by 51% between 2000 and 2010 (see table 1). Populations in remote areas have more limited options for managing land and accessing other benefits of economic development.⁽²⁾

Improving land quality and living standards of the rural population requires policy responses that improve the condition of terrestrial ecosystems by avoiding, reducing and reversing degraded land. Investments, particularly in hotspot locations characterized by both high restoration potential and high socioeconomic benefits in poverty areas, will improve the conditions of the most vulnerable people and increase the resilience of ecosystems.

Table 1: Population on degrading agricultural land in Malawi^(3,4,5)

Population categories ⁱ	2000	2010	% change from 2000 to 2010
Rural population on degrading agricultural land	862,496	1,322,994	53.4%
Share (%) of rural population on degrading agricultural land	9.0%	10.6%	*1.6%
Rural population on remote degrading agricultural land	136,540	206,792	51.5%
Share (%) of rural population on remote degrading agricultural land	1.4%	1.7%	*0.2%
Rural population	9,557,899	12,474,593	30.5%
Total population	11,193,230	14,769,824	32.0%

* Note: Due to rounding, some figures in the text may not correspond with those reported in the tables or the sum of separate figures. * percentage-point difference between 2000-2010.

2.2 Regional and Global Overview

In Africa, 23% of the continent's rural population resided on degrading agricultural land in 2010, equivalent to 184 million people. Moreover, 6% of the total rural population – or 47.6 million people – lived in remote degrading agricultural land with limited access to markets.

The changes in these indicators between the period 2000 and 2010 for the region depict increases of 35% and 38% for the case of population residing in degrading agricultural land and remote degrading agricultural land respectively; whereas the overall population in rural areas grew at 27% over the same period (see table 2).

On a global level, it is estimated that about 1.5 billion people worldwide – equivalent to 32% of the total rural population – resided on degrading agricultural land in 2010. Furthermore, during the same year, 233 million people lived on remote degrading agricultural land with limited access to markets, representing 5% of the global rural population.

Among the world's regions suffering from land degradation, the most affected continent is Asia with 79% of the global rural population residing in degrading agricultural areas (or 1.1 billion people). The second most affected region is Africa, with a share of 12% in the global rural population living in degrading agricultural areas. The remaining 9% are spread across Europe (5%), Latin America and the Caribbean (3%), and Northern America and Oceania (1%).

Regarding changes over time for the period 2000-2010, the global rural population in degrading agricultural areas and remote degrading agricultural areas increased by 12% and 14%, respectively.

Table 2 contains additional details of the populations living in degrading agricultural areas and remote degrading agricultural areas by region and globally for the years 2000 and 2010, as well as the percentage changes during this decade.

Table 2: Population on degrading agricultural land at regionalⁱⁱⁱ and global scale⁽³⁾

Regions	Population in 2010					% change from 2000 to 2010		
	Rural population (in millions)	Rural population on DAL (in millions)	% share	Rural population on remote DAL (in millions)	% share	Rural population	Rural population on DAL	Rural population on remote DAL
Africa	812.6	184.0	22.6%	47.6	5.9%	26.8%	34.7%	37.6%
Asia	3,102.9	1,176.8	37.9%	175.0	5.6%	12.2%	10.9%	8.6%
Europe	310.1	75.6	24.4%	2.5	0.8%	-2.6%	-6.5%	-5.9%
Latin America and the Caribbean	350.9	48.2	13.7%	7.2	2.1%	14.1%	17.8%	16.4%
Northern America	71.4	11.4	16.0%	0.7	1.0%	7.5%	7.5%	6.2%
Oceania	16.0	0.9	5.4%	0.3	1.8%	15.0%	0.8%	39.3%
World Total	4,663.9	1,496.9	32.1%	233.3	5.0%	13.4%	12.4%	13.6%

Note: DAL= Degrading Agricultural Land

3. Economics of Land Degradation

3.1 National Overview

Land provides valuable ecosystem services for human well-being, but land degradation leads to a reduction in the provision of these services with significant social and economic costs to the country. The decline of ecosystem services can take different forms, including decline in food availability, soil fertility, carbon sequestration capacity, wood production, groundwater recharge, among others.^(6, 7, 9)

The costs of land degradation for the country are measured in terms of the changes in land productivity by considering two aspects: changes in land cover from a high-value biome to a lower-value biome (e.g. forest land converted to cropland); and the decline in ecosystem services provision within a certain land cover type due to degrading land-use practices (e.g. reduced cropland productivity over time).⁽⁶⁾

In Malawi, the total annual cost of land degradation is estimated at 320 million United States Dollars (USD) — this is equal to 7% of the country's Gross Domestic Product (GDP).^{iv} Moreover, a considerable share of the costs of land degradation (53%) is due to the decline in provisioning ecosystem services (e.g. food availability, wood production, etc.), which has a significant impact on the population of the country. The remaining share refers to the regulating ecosystem services (e.g. carbon sequestration, water regulation flows), which has an impact not only at the country level, but also on the regional and global scale due to the transboundary nature of these services that provide incentives for international cooperation (see table 3).^v

Land degradation often stems from land-use decision-making processes driven by high market prices of specific ecosystem services — for example, food. In this context, land-use decisions may largely neglect the significance of other ecosystem services for which no markets exist, but which are also of high value to the society.⁽⁹⁾

Given the significant economic burden of land degradation, research has also focused on the study of the costs of action against land degradation through restoration and sustainable land management practices. These costs of action are often compared to the costs of inaction — the latter being derived from the projection of past degradation rates to the future.

In this context, a recent global assessment on land degradation⁽⁶⁾ shows that for Malawi the returns on taking action against land degradation versus inaction are estimated at 3 USD for every dollar invested in reverting degraded land,^{vi} underlining the strong economic incentives for bold actions on achieving LDN.

Table 3: Economics of land degradation (LD) in Malawi^(4, 6, 8)

Total annual cost of land degradation (base year 2007)	320 m USD
Cost of LD due to the decline in provisioning ecosystem services (as % of total cost)	53%
Cost of land degradation as % of GDP	7%
Cost of action (30-year planning horizon)	4 bn
Cost of inaction (30-year planning horizon)	14 bn
Returns on action against land degradation per dollar invested	3 USD
GDP 2016 (USD)	5.4 bn
Share of Agriculture in total GDP 2016	28%
GDP per capita 2016 (USD)	301

Note: m = million; bn=billion

3.2 Regional and Global Overview

For Africa, the total annual costs of land degradation are estimated at 65 billion USD, which amounts to about 4% of the total GDP of the region. This share, however, varies considerably among countries.

On a global scale, the costs of land degradation are estimated at about 297 billion USD.^{vii} As illustrated in table 4, Asia accounts for the largest share of the total global cost of land degradation (28%), followed by Africa (22%), Latin America and the Caribbean (20%), Northern America (12%), Europe (12%) and Oceania (5%).

Assessments of the cost of action against land degradation versus the cost of inaction show that the latter significantly outweighs the former. On the regional level, the costs of action for Africa are estimated at 731 billion USD, whereas the costs of inaction equal about 3.1 trillion USD^{vi} (see table 4). The regional breakdown reveals social returns ranging from about 4 USD in the case of Asia, Africa, and Latin America and the Caribbean, and up to 6 USD in Europe, Northern America, and Oceania (see table 4).

On a global level, estimates show costs of action in the amount of 4.6 trillion USD, whereas the costs of inaction equal about 23.2 trillion USD.^{vi} That means that the expected social returns of taking action are estimated at about 5 USD for every dollar invested in the restoration of degraded land and sustainable land management.

Table 4: Cost of land degradation at regionalⁱⁱⁱ and global scale⁽⁶⁾

Regions	Cost of Land Degradation (LD)		Cost of action and inaction		
	Total annual cost of LD (in billion USD; year 2007)	% of the annual cost of LD in the world total	Cost of action in 30-year time horizon (in billion USD)	Cost of inaction in 30-year time horizon (in billion USD)	Returns on action against LD (in USD)
Africa	65	22.0	731	3,112	4
Asia	84	28.4	976	4,359	4
Europe	35	11.8	945	5,652	6
Latin America and the Caribbean	61	20.4	789	3,107	4
Northern America	36	12.2	759	4,599	6
Oceania	15	5.2	407	2,442	6
World Total	297	100.0	4,606	23,272	5



4. Land and Climate Change

Land plays an important role in the global carbon cycle because terrestrial ecosystems continuously exchange carbon fluxes with the atmosphere. The exchange is two-way: on the one hand, terrestrial ecosystems sequester carbon through natural processes, and on the other hand, they release carbon through respiration as well as anthropogenic activities related to agriculture, forestry, and other land use. The role of terrestrial ecosystems as a source and sink of emissions positions land as a key element of intervention for climate change mitigation and adaptation.

4.1 National Overview

Land as a Source of Emissions

The Agriculture, Forestry and Other Land Use (AFOLU) sector is an important source of Greenhouse Gases (GHG). Figures vary on how this sector contributes to the national emission inventories across countries. In Malawi, the AFOLU sector is responsible for 78% of the total emissions of the country (see table 5).

Within Malawi's AFOLU sector, the larger share of the emissions is from Forestry and Other Land Use (FOLU) subsector (54%). Emissions from Agriculture plays a minor role.

Land as a Carbon Sink

Terrestrial ecosystems also play an important role as carbon sinks, offsetting emissions released by various sectors of the economy. The removals of carbon emissions through Forest are estimated at 3 million tonnes of CO₂ in 2010 for Malawi (see table 5). This is equal to 16% of the total emissions of the country. The potential carbon storage per hectare (ha) and year varies considerably depending on the type of biome, the practice on the ground, and the prevalent climate.⁽¹¹⁾ The mean rate of sequestration is estimated at 1.5 tonnes of carbon (tC)/ha per year, where 0.5 tC is from soil organic carbon sequestration and an additional 1.0 tC from biomass.^{viii (11)}

In general, terrestrial ecosystems have a significant potential for carbon sequestration linked to the cumulative historic loss of carbon from land-use change. The capacity of land to further store carbon is crucial for bridging the time until new technologies to tackle climate change are adopted on a larger scale.⁽¹¹⁾

Table 5: Land as a source and sink of emissions⁽¹⁰⁾ in Malawi (year 2010)

Sectors	GHG (Mt-CO ₂ e)	%
Sources total	17.83	100.0
AFOLU sources	13.94	78.2
Agriculture	4.26	23.9
FOLU net sources	9.67	54.3
Other sectors	3.89	21.8
Forest net sink	-2.86	
FOLU total	6.82	
Total net emissions with FOLU	14.97	
Total net emissions per capita with FOLU (in tonnes of CO ₂ e)	1.0	

Note: GHG=Greenhouse Gases; Mt-CO₂e = million tonnes of carbon dioxide equivalent.

The UNCCD Science Policy Interface developed the Land Degradation Neutrality (LDN) conceptual framework⁽¹²⁾, which refers to three hierarchical policy responses to achieve LDN that go hand in hand with climate actions: i) **avoid** further land degradation by halting conversion of land types, for example, not converting forest land into agricultural land; ii) **reduce** the impact of land-intensive activities by using Sustainable Land Management (SLM) practices, so that less carbon is released from soil, crops and other biomass; and iii) **reverse** land degradation, for example, by restoring or rehabilitating land that has lost productivity.⁽¹²⁾

Land as a cost-effective mitigation option

Within the various climate change mitigation alternatives, land-based mitigation options rank among the most cost-effective opportunities to sequester or avoid carbon.⁽¹³⁾ The cost of capturing one tonne of carbon (tC) by restoring degraded land is estimated at 51 USD per tC; while alternative engineering techniques such as 'gas plant capture and carbon sequestration' have a cost of 306 USD per tC (see table 6). Moreover, land-based mitigation options are estimated to be more cost-effective than other widely-used strategies to avoid emissions — for example, the substitution of fossil fuels by solar or wind energy.^(13, 14)

Moreover, it is worth noting that the option of storing carbon in terrestrial ecosystems by restoring land generates several other co-benefits that should also be factored in. They include for instance improving soil health, reducing food insecurity and enhancing water regulation flows.

Table 6: Cost of carbon sequestration using different techniques^{ix(13)(14)}

Technique/Strategy	Cost of abatement USD per tC
Second-generation biofuels	25
Pastureland afforestation	51
Degraded-land restoration	51
Degraded forest restoration	61
Agriculture conversion	128
Biomass co-firing power plant	153
Coal-C capture and sequestration	229
Gas plant capture and sequestration	306
Solar VP*	92
Wind*	76

Note: tC= tonne of Carbon

Land matters play a key role in developing climate change mitigation and adaptation policies. The following box presents the leading land-based mitigation and adaptation strategies considered in Malawi's Nationally Determined Contributions.



Box. Highlights on Climate Change and Land from Malawi's Nationally Determined Contributions⁽¹⁵⁾

Land-based mitigation plans

Agriculture, Forestry and Other Land Uses (AFOLU) target: -0.4 MtCO₂e per year by 2040; -2.6 MtCO₂e per year

Forestry and land use: Afforestation, reforestation and forest conservation and protection of catchments; upscale afforestation, reforestation and forest conservation and protection of catchments; 2% increase in forest cover nationally; implementation of payment for ecosystem service for hydroelectric dams; promotion of non-extractive livelihoods from forest; upscale the non-extractive livelihoods from forest; promote and upscale sustainable production of fuel wood by establishing woodlots plantations and forest management

Conservation: Protection and conservation of protected areas

Agriculture: Market-based policies and legal instruments; Support development of market based policies and legal instruments to shift decisions from financial to environmental decisions; develop appropriate extension and training materials for climate resilient agronomic practices; upscale the dissemination of climate resilient agronomic practices to above 10% of current cropland; build capacity to implement and monitor the agriculture NAMA

Fertilizer: Reducing synthetic fertilizer application

Agricultural Waste: Composting manure

Land-based adaptation priorities

Afforestation: Support research in drought tolerant and fast growing tree species; Expand afforestation and forest regeneration programmes

Land and Soil Management: Promote improved land use practices

Agroforestry: Implement conservation agriculture and agroforestry practices

Irrigation: Increase irrigation at smallholder level; Increase land under irrigation through Greenbelt initiative from 20,000 to 40,000 ha; Promote on-farm water conservation technologies; Expanded programmes of Greenbelt initiative from 40,000 ha to 100,000 ha by 2030; Support an expanded programme of constructing multipurpose dams for irrigation and aquaculture

Crops: Promote the growing of drought tolerant crop varieties

Disaster Risk Management (DRM): Build adaptation capacity in climate resilient agronomic practices for smallholder farmers; Develop financial mechanisms to support crop insurance targeting smallholder farmers

Ecosystem and Biodiversity: Provide watering points at strategic locations of national park/ game reserve; Implement diseases control programmes; Support capacity building in wildlife institution to lead in adaptation initiatives e.g. translocation and culling

Water Management: Promote water harvesting technologies at all levels; Implement integrated catchment conservation and management programme; Support an expanded programme of constructing multipurpose dams to enhance water storage

Biomass Energy: Promote use of biomass briquettes as substitute for firewood and charcoal; Promote an energy mix that moves people away from use of biomass; Support an expanded programme of briquette production and use

Transport: Construct infrastructure for flood control, transport, etc.

Mitigation and adaptation policy frameworks: REDD+ programme action plan; National Environmental Action Plan (NEAP); National Environmental Policy

4.2 Regional and Global Overview^{xi}

In Africa, 63% of the total emissions released were from the Agriculture, Forestry and Other Land Use (AFOLU) sector in the year 2010. This percentage represents 2,610 Mt-CO₂e out of the total 4,109 Mt-CO₂e emitted in the region (see table 7). In the AFOLU sector, the 'Forestry and Other Land Use (FOLU)' subsector accounts for 44% (or 1,816 Mt-CO₂e), while the 'Agriculture' subsector is responsible for 19% (or 794 Mt-CO₂e) of the total emissions from the region.

At a global level, it is estimated that the AFOLU sector is responsible for 23% of the GHG emissions, which is equal to 11,380 Mt-CO₂e (see table 7). Breaking down the AFOLU sector into 'Agriculture' and 'FOLU' shows that the majority of emissions come from the latter subsector with a total amount of 6,304 Mt-CO₂e; while Agriculture emitted 5,075 Mt-CO₂e.

Regarding the regional contributions to the global emissions of the AFOLU sector, greenhouse gas (GHG) inventories report that the Asia region is the leading contributor of global AFOLU emissions. Asia is responsible for 35% of global AFOLU emissions, followed by Latin America and Africa which are responsible for 24% and 23% of emissions respectively. Table 7 displays further details of the regional contributions of the AFOLU sector in relation to the total global emissions as well as the regional breakdown for the Agriculture and FOLU subsectors.

Evidence also shows that the global forest ecosystems alone removed 3,234 Mt-CO₂e from the atmosphere in the year 2010 (see table 7). More generally, out of the total global carbon emissions to the atmosphere by human activities, an estimated 42% are accumulated in the atmosphere; another 23% is sequestered by the oceans; and the remaining 34% is attributed to sequestration by terrestrial ecosystems,⁽¹¹⁾ highlighting the essential role of land-based ecosystems to mitigate climate change.

Table 7: Regionalⁱⁱⁱ and global emissions/removals from the Agriculture, Forestry and Other Land Use (AFOLU) sector and related indicators⁽¹⁰⁾ in 2010

Regions	Sources total		AFOLU Net sources		Agriculture		FOLU net sources		Forest net sink		Total emissions per capita with FOLU Mt-CO ₂ e
	Mt- CO ₂ e	in %	Mt- CO ₂ e	in %	Mt- CO ₂ e	in %	Mt- CO ₂ e	in %	Mt- CO ₂ e	in %	
Africa	4,109	8.3	2,610	22.9	794	15.7	1,816	28.8	-159	4.9	3.8
Asia	23,421	47.5	3,974	34.9	2,262	44.6	1,712	27.2	-936	28.9	5.4
Europe	8,268	16.8	875	7.7	567	11.2	308	4.9	-847	26.2	10.1
Latin America and the Caribbean	4,838	9.8	2,724	23.9	896	17.7	1,828	29.0	-545	16.9	7.2
Northern America	7,711	15.6	752	6.6	406	8.0	346	5.5	-494	15.3	21.0
Oceania	1,001	2.0	445	3.9	150	3.0	295	4.7	-253	7.8	20.7
World total	49,349	100	11,380	100	5,075	100	6,304	100	-3,234	100	6.7

Note: Mt-CO₂e = million tonnes of carbon dioxide equivalent; FOLU= Forestry and Other Land Use.

5. Opportunities – The Way Forward

The 2030 Agenda for Sustainable Development offers opportunities for countries to curb the growing threats of land degradation and to reap multiple socioeconomic benefits of LDN.

Sustainable Development Goal 15 'Life on Land' and its target 15.3 on Land Degradation Neutrality (LDN) particularly encourage countries to 'combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land degradation-neutral world by 2030'.

In October 2015, UNCCD country Parties decided that striving to achieve SDG target 15.3 is a strong vehicle for driving the implementation of the Convention and requested the UNCCD secretariat and appropriate UNCCD bodies to take the initiative and invite other relevant agencies and stakeholders to cooperate on achieving SDG target 15.3 (decision 3/COP12).

To achieve SDG target 15.3, the following five elements have been identified:

- (1) **LDN targets:** setting targets and establishing the level of ambition;
- (2) **Leverage and impact:** catalyzing the multiple benefits that LDN provides from climate change mitigation and adaptation to poverty reduction;
- (3) **Partnerships and resource mobilization:** rationalizing engagement with partners, overcoming fragmentation and systematically tapping into increasing finance opportunities, including climate finance;
- (4) **Transformative action:** designing and implementing bold LDN transformative projects that deliver multiple benefits; and
- (5) **Monitoring and reporting:** tracking progress towards achieving the LDN targets.

As of June 2018, 118 countries have made the commitment to translate the global goal of achieving LDN by 2030 into national action by setting national voluntary targets with the support of the LDN Target Setting Programme (LDN TSP) – a programme established by the Global Mechanism in collaboration with the UNCCD secretariat and supported by various partners. Malawi is among the countries that have already set the national voluntary LDN target, established the LDN baseline, and formulated associated measures.

The LDN targets provide Malawi with a strong vehicle for fostering coherence of policies and actions by aligning the national LDN targets with measures from the Nationally Determined Contributions and other national commitments, such as the restoration of 4.5 million hectares of degraded land under the Bonn Challenge and AFR100 initiatives.

Investing in LDN also accelerates the advancement of other SDGs due to the close linkages between land and other goals and targets, such as: Goal 1 (No poverty), Goal 2 (Zero hunger), Goal 5 (Promote gender equality), Goal 6 (Clean water and sanitation), Goal 8 (Decent work and economic growth), and Goal 13 (Climate action).⁽¹⁶⁾

Box 2. Malawi's National Voluntary LDN Targets and Measures

At national scale Malawi has set a target of achieving land degradation neutrality by 2030 as compared to the baseline of 2015 (no net loss) and an additional 2% of the land territory of 9.4 million hectares improvement. This translates to 188,000 hectares with net gain as compared to 2015.

Sub-regional level the targets:

- LDN is achieved in the High Lands of Nyika, Viphya and Mulanje, Dedza and Zomba mountains by 2030 as compared to 2015 (no net loss)
- LDN is achieved in the land degradation Hotspots along the Rift Valley Escarpment Area of Malawi by 2030 as compared to 2015
- Attain land degradation neutrality on the Plateaux ecological zone by 2030 as compared to 2015
- Attain land degradation neutrality in the Shire River basin catchment by 2030 compared to 2015 and an additional 2% of the basin has improved (Net gain)

Specific targets for avoiding, minimizing and reversing land degradation:

- i. Improve productivity of 754,320 hectares cropland by 2030
- ii. Improve Soil Organic Carbon (SOC) stocks on cropland to 55 ton/ha by 2025 as compared to 44.7 ton/ha estimated in 2015
- iii. Rehabilitate one million hectares of degraded land for crop production by 2030
- iv. Halt the conversion of forests and wetlands to other land cover classes by 2020
- v. Improve forest (plantation & indigenous) cover by 33,750 hectares by 2030 as compared to 2015
- vi. Reduce the rate of top soil loss (soil erosion) to 20 tons per hectare per year by 2030 from the 2015 estimated rated of 29 tons/ha/year
- vii. Increase forest cover by 2% from 2015 baseline by 2022³
- viii. Restore 820,000 hectares of degraded indigenous forest by 2030³
- ix. Sustainably manage 138,000 hectares of plantation forest by 2025³
- x. Restore 36,000 hectares of degraded stream banks by 2030³
- xi. Protect 2.4 million hectares of natural forest by 2035

Source: Ministry for Natural Resources, Energy and Mining Department of Forestry,
Republic of Malawi

6. Ongoing Projects and Programmes

To illustrate land-based approaches, the following section features some of the ongoing projects and programmes supported by national and international organizations.^{xiii}

Shire Natural Ecosystems Management Project. The objective of project is to improve land and water management for ecosystem and livelihood benefits in target areas. Implementing Agency: The World Bank. GEF Grant/Cofinancing: 6.5 million USD/72.7 million USD. Link: for further information click [here](#).

Enhancing the Resilience of Agro-Ecological Systems. The project's goal is to enhance the Provision of Ecosystem services and improve the Productivity and Resilience of Agricultural Systems of Vulnerable Rural Poor. Implementing Agency: International Fund for Agricultural Development. GEF Grant/Cofinancing: 7.1 million USD/ 87.3 USD. Link: for further information click [here](#).

Enhancing Sustainability of Protected Area Systems and Stabilizing Agro-production in Adjoining Areas through Improved IAS Management. This project aim to prevent new invasions and reduce the current impacts of invasive alien species in protected areas and adjoining agro-ecosystems in Malawi. Implementing Agency: United Nations Environment Programme. GEF Grant/Cofinancing: 1.5 million/5.1 million USD. Link: for further information click [here](#).

Effective Management of Nkhosakota Wildlife Reserve. The objective of project is to ensure effective management of the Nkhosakota Wildlife Reserve through a sustainable management model focusing on its Bua watershed area. Implementing: The World Bank. GEF Grant/Cofinancing: 845,000 USD/2.4 million USD. Link: for further information click [here](#).



7. Country Studies

For further reading, this section offers country studies that may be useful in making the case for investing in Land Degradation Neutrality.

1. **Economics of Land Degradation and Improvement in Tanzania and Malawi.** — Kirui, O. (2015).
2. **The Economic Linkages between Rural Poverty and Land Degradation: Some Evidence from Africa.** — Barbier, E. (2000).
3. **Enhancing Adaption of Forests and People in Africa - Development of Pilot Cases for Selected Forest Ecosystems in Ghana and Malawi.** — Foli. et al. (2011).
4. **Yield Response of Maize (Zea Mays L.) to Conservation Agriculture Cropping System in Southern Africa.** — Thierfelder. et al. (2014).
5. **Economic Aspects of Community Involvement in Sustainable Forest Management in Eastern and Southern Africa.** — Mogaka, H. et al. (2001).



8. Supplementary Information

8.1 Glossary

This subsection provides a brief description of the indicators presented above.

Annual cost of land degradation

The UNCCD defines land degradation as 'any reduction or loss in the biological or economic productive capacity of the land resource base. It is generally caused by human activities, exacerbated by natural processes and often magnified by and closely intertwined with climate change and biodiversity loss.' In the study featured here on the cost of land degradation, Nkonya and colleagues⁽⁶⁾ approach the study of land degradation by investigating declines in land productivity in the past due to: i) land cover changes from a high value-biome to a lower-value biome, such as the conversion from forest land into cropland; and ii) declines in the ecosystem services provision within a land cover type due to the use of degrading practices.

Cost of action

The costs of action are estimated by taking into account the following two cost categories: i) initial fixed investments and maintenance expenses that are related to the restoration of the high-value biome until it reaches biological maturity; ii) the inclusion of the opportunity cost given by the forgone benefits from the lower-value biome under replacement. The analysis of the cost is carried out over a planning period of 30 years.⁽⁶⁾

Cost of inaction

Cost of inaction represents the 'business as usual' (BAU) scenario. In this case, future land degradation trends are assumed to continue along patterns similar to those of the past. The total costs of inaction are calculated by the sum of future annual costs of land degradation over a 30-year planning horizon - where land degradation is captured by land cover changes from a high-value biome to a lower-value biome.⁽⁶⁾

Returns of action

Nkonya and colleagues⁽⁶⁾ measure the benefit of action as the difference between the cost of inaction minus the cost of action. When this difference is positive, then taking action is justified in economic terms. Moreover, the figures on returns on investment are calculated as the cost of inaction over the cost of action. For further methodological details on the annual cost of land degradation, cost action, inaction and returns on action, see Nkonya and colleagues.⁽⁶⁾

Population on degrading agricultural land

Estimates of the population in degrading agricultural areas are based on the work of Barbier and Hochard.⁽³⁾ They identify agricultural degrading land by looking at the areas that experienced negative changes in net primary productivity, using the Normalized Difference Vegetation Index. Note that estimates are mainly constrained to populations residing on 'agricultural land' in this study; the consideration of other land cover types may therefore increase the magnitude of these figures. Regarding data on the spatial distribution of rural population, this study uses data published by the Global Rural-Urban Mapping of the Socioeconomic Data and Applications Center (SEDAC). In order to further identify population in remote areas, Barbier and Hochard⁽³⁾ use data from the Global Environment Monitoring Unit of the Joint Research Centre of the European Commission.

Rural poverty

The rural poverty headcount ratio is used to calculate rural poverty, i.e. the percentage of rural population living below the national poverty line. National poverty line is the benchmark for estimating poverty indicators that are consistent with the country's specific economic and social circumstances and reflect local perceptions of the level and composition of consumption or income needed to be non-poor.⁽⁴⁾

Sustainable Land Management

SLM is the use and management of land resources—soil, water, animals and plants – for the production of goods to meet changing human needs, while ensuring the long-term productive potential of these resources and the maintenance of environmental functions. Degradation of water, soil and vegetation as well as emissions contributing to climate change can be limited through SLM practices that simultaneously conserve natural resources and increase yields.

8.2 Notes

- i. Figures on population on degrading agricultural land are calculated by using the shares of rural population on degrading agricultural land and remote degrading agricultural land estimated in the work of Barbier and Hochard⁽³⁾, in combination with data on rural population from the World Bank Development Indicators.⁽⁴⁾
- ii. Population in remote degrading areas is identified in terms of market accessibility, where access to market is defined as less than five hours of travel to a market city with a population of 50,000 or more.⁽³⁾
- iii. Country grouping is based on geographic regions as defined by the United Nations Statistics Division (see: <https://unstats.un.org/unsd/methodology/m49/>.)
- iv. Estimates of the economic costs of land degradation illustrated in this country profile are based on the work of Nkonya and colleagues.⁽⁶⁾
- v. The relationship between food production (provisioning ecosystem service) and the supply of other ecosystem services often depicts important trade-offs.⁽¹⁾
- vi. These figures correspond to a 30-year planning horizon in terms of quantification of costs and benefits.
- vii. Global estimates of the costs of land degradation vary to a great extent depending on the study. A study led by the Economics of Land Degradation Initiative⁽⁹⁾ estimates the global costs of land degradation at 9.6 trillion USD. In this regard, the figures presented in the current publication are conservative.
- viii. This is a global average coefficient used as a default in this publication, and it should be replaced with that of national level when available. Note also that one tonne of carbon (C) is approximately equivalent to 3.66 tonnes of carbon dioxide (CO₂).
- ix. This version of the country profile uses the 'Global GHG Abatement Cost Curve' as default information.⁽¹⁴⁾ National GHG Abatement Cost Curve should be used when available.
- x. Although solar and wind power are not sequestration techniques, but rather technologies that avoid (or reduce) emissions from the source, figures still show how competitive is restoring degraded land in comparison with solar or wind abatement alternatives.
- xi. Figures related to Greenhouse Gases in this subsection are retrieved from FAOSTAT.⁽¹⁰⁾
- xii. The information on projects and programmes presented in this section has been obtained from the websites of the following organizations: Climate Investment Funds, Food and Agriculture Organization of the United Nations, Global Environment Facility, Green Climate Fund, United Nations Development Programme, United Nations Environment Programme and the World Bank.

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8.4 Photos

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8.5 About this Publication

This country profile is intended to provide a brief overview of recent studies, assessments and indicators that demonstrate multiple benefits of taking bold actions to achieve Land Degradation Neutrality.

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